PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

GRAB CRANE 抓斗起重机(4311-GCR-001 - 002)

MTI-CHP-CPM-SOP-001_Rev. 000

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the grab crane under complete closure(4311-GCR-001/002).

此程序涵盖操作员对完全关闭下的抓斗起重机进行目视操作前检查、启动运行和停车的责任(4311-GCR-001/002)。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the grab crane is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the grab crane and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对抓斗起重机进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定抓斗起重机的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures. MTI-CP-SOP岗位操作规程
- Grab Crane(4311-GCR-001/002). 抓斗起重机(4311-GCR-001/002)

4. PPE 个人保护装备



5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue危害	Control Strategy控制
1. Mechanical injury. 2. Dust. 3. Electric shock. 4. Slippery conditions and a trip to the danger. 5. High falling objects. 6. Vehicle collision crush. 1. 机械伤害; 2. 灰尘; 3. 触电;	1. Wear protective equipment according to the requirements, and no standing people or pedestrians under the sling. 2. Wear dust mask as required. 3. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective. 4. Please hold the handrail up and down the stairs.

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Description of Hazard / Environmental Issue危害	Control Strategy控制
4. 湿滑的条件，绊倒危险； 5. 高空落物； 6. 车辆碰撞碾压。	5. Stay away from the crane walking area. 6. Walk along the prescribed route and stay away from the cooked pellet transport vehicles. 1. 按要求穿戴好防护用品，吊物下严禁站人或行人； 2. 按要求佩戴防尘口罩； 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 4. 上下楼梯请扶好扶手； 5. 远离起重机行走区域； 6. 按照规定的路线行走，远离铁球团运输车辆。

6. Equipment 工具/ 设备

4311-MOT-001/005	Lifting motor起升电机
4311-MOT-002/006	Opening and closing motor开闭电机
4311-MOT-003/007	Small traveling motor小车运行电机
4311-MOT-004/008	Big traveling motor大车运行电机
	Clean the site tools清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training and obtain the qualification certificate. 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site. 3. Check and clean the equipment and the site sanitation. 4. Contact with the process before and after and give clear instructions. 1. 确认经过岗前技术培训，并取得相应的资格证书； 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患； 3. 检查并打扫设备、现场卫生； 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Grab Crane(4311-GCR-	1. Check the lighting device to ensure that the illumination and power supply voltage meet the

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№	Stage	Key Points
	001/002) 检查抓斗起重机 (4311-GCR-001/002)	corresponding requirements; 2. Check whether the main girder, end beam, and trolley frame of the crane are normal; 3. Check the tightening of the stressed components, motors, speed reducer, brakes, coupling and other connecting bolts; 4. Check whether the grab pin shaft is loose and detached, and whether the grab pin is opened and closed freely; 5. Check whether the parts of the brake have cracks, plastic deformation, excessive wear and other defects, the hydraulic brake has no oil leakage; the brake adjustment is appropriate, the brake is stable and reliable; the brake wheel and the friction plate are open, and the brake contacts the friction plate evenly when the brake is closed; 6. Check that the wire rope should not be broken with excessive breakage, excessive wear and serious corrosion, and the wire rope on the drum is neatly arranged, without job-hopping and rope pressing; the wire rope should not be bent or flattened; the end of contact wire rope should meet the corresponding requirements, Whether the lubricating oil is in good; 7. Check whether the drum has cracks, excessive or abnormal wear; 8. Check whether the pulley has cracks, wheel rim damage, etc. The pulley should rotate flexibly, the pulley anti-rope removal device is effective, and the stop screw on both sides of the cover is not loose; 9. Check that the reducer has no abnormal sound, no vibration, no oil leakage; 10. Check the coupling connection without movement, running without sound; 11. Check that the clearance and height difference of the cart track meet the corresponding requirements; 12. Check the suspension part of the driver's cab for no unwelding, crack and no loosening phenomenon; 13. Check that the alarm bell works normally, the loudness is suitable, and the indication signal is clear; 14. Check the operation of each mechanism is flexible, no blocking, zero feel is clear, the insulation protection of the control lever is not damaged, the emergency stop button can not be automatically reset, the emergency stop switch can cut off the power supply of the crane; 15. Check that the limit position limiter of each running direction is flexible and reliable, and the lifting weight limiter and overspeed protection device are complete

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№	Stage	Key Points
		and effective; 16. Check the normal reliability of the control cabinet and electrical facilities. 1. 检查照明装置，确保照度及供电电压符合相应要求； 2. 检查起重机主梁、端梁、小车架等状况是否正常； 3. 检查受力构件、电机、减速箱、制动器、联轴器等连接部分螺栓紧固； 4. 检查抓斗销轴是否松动、脱出，抓斗是否开闭自如； 5. 检查制动器的零部件是否有裂纹、塑性变形、过度磨损等缺陷，液压制动器无漏油；制动器调整适宜，制动平稳可靠；制动器打开时制动轮与摩擦片无摩擦现象，制动器闭合时制动器与摩擦片接触均匀； 6. 检查钢丝绳不应有断丝超标、磨损过度及严重腐蚀等现象，卷筒上钢丝绳编排整齐，没有跳槽和压绳等现象；钢丝绳不应有折弯、压扁现象；接触钢丝绳端部固定情况，应满足相应要求；润滑脂是否良好； 7. 检查滚筒是否有裂纹、过度或异常磨损； 8. 检查滑轮是否有裂纹、轮缘破损等，滑轮应转动灵活，滑轮防脱绳装置有效，外罩两侧止退螺帽无松动； 9. 检查减速机无异常声响，无振动，无漏油； 10. 检查联轴器连接无窜动，运转无声音； 11. 检查大车轨道间隙、高低差符合相应要求； 12. 检查司机室悬挂部分无脱焊、裂纹，无松动现象； 13. 检查警铃工作正常，响度适合，指示信号清晰； 14. 检查各个机构操纵灵活、无卡阻，零位手感明确，操纵杆绝缘保护无破损，紧急停止按钮不可自动复位，急停开关能够切断起重机动力电源； 15. 检查各个运行方向的极限位置限位器灵活可靠，起重量限制器、超速保护装置齐全有效； 16. 检查控制柜及电气设施正常可靠。

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№	Stage	Key Points
3	Check the other Equipment and Facilities 检查其它设备设施	1. Motor Check whether the lifting motor (4311-MOT-001/005), opening and closing motor (4311-MOT-002/006), small traveling motor (4311-MOT-003/007) and big traveling motor(4311-MOT-004/008) are grounded intact and the rotation direction is correct. 1. 电机 检查起升电机(4311-MOT-001/005)、开闭电机(4311-MOT-002/006)、小车运行电机(4311-MOT-003/007)、大车运行电机(4311-MOT-004/008)接地是否完好，旋转方向是否正确。
4	Inspect Materials Level 检查物料料位	1. Check whether the lifting grab bucket is buried deep in the ripe pellet; 2. Check whether the transport vehicle is in the center of the loadometer (***) and whether there are ripe pellets scattered around the loadometer (***). 1. 检查起重抓斗是否深埋铁球团中； 2. 检查铁球团运输车辆是否在地磅(***)中央，地磅(***)四周有无铁球团散落。
5	Start-up of Grab Crane(4311-GCR-001/002) 启动抓斗起重机(4311-GCR-001/002)	Implement the driving "ten not hanging" system ⟹ Drive power supply, lifting motor (4311-MOT-001/005), open and closed motor (4311-MOT-002/006), trolley running motor (4311-MOT-003/007), cart running motor (4311-MOT-004/008) are all charged, buttons and levrods ⟹ Start from the grab opened and placed on the material, due to the action of gravity, the sliding block drops, and the convex block is at about 30 angles by the thimble ⟹ Lift the wire rope, the whole slide moves up along the guide rail, move a small distance between the bump and the groove (the bump rotates again at a small Angle), drive the whole lower girder to move up, complete the grab closing and grasp the material ⟹ Click the alarm bell to prompt, and then grab crane (4311-GCR-001/002) march above the bucket of the loadometer (***) ⟹ When unloading, the wire rope is released. At this time, the bottom of the bucket contacts the support surface first. Due to the action of gravity, the sliding block drops, and the convex block is headed horizontally by the thimble ⟹ Lift the wire rope, slide block upward, the cam is blocked by the groove to the vertical, on both sides of the guide rail, so the sliding seat and the lower beam separation, upward to the upper beam, the grab left the ground immediately due to weight, the bucket open and unloading ⟹ The unloaded grab is open in the air,

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№	Stage	Key Points
		placed on the material, the sliding block drops, and the cam is reached at about 30 angles by the thimble to complete a grasping process ⟶ The grab should not be suspended in the air, During work stoppage and rest. 执行行车“十不吊”制度⟶驱动电源送电，起升电机(4311-MOT-001/005)、开闭电机(4311-MOT-002/006)、小车运行电机(4311-MOT-003/007)、大车运行电机(4311-MOT-004/008)全部带电，各种按钮和各种操纵杆带电⟶先把抓斗打开放在物料上面开始，由于重力作用，滑移块下降，凸块被顶针顶成约30°角⟶提钢丝绳，整个滑移块顺着导轨上移，移动一小块距离之后凸块与凹槽衔接（凸块再次小角度旋转），带动整个下承梁上移，完成抓斗关闭抓料⟶点按警铃进行提示，然后抓斗起重机(4311-GCR-001/002)行进到地磅(***)上的铁球团运输车辆车斗上方⟶卸料时，释放钢丝绳，此时抓斗底部先接触支撑面，由于重力作用，滑移块下降，凸块被顶针顶成水平方向⟶提钢丝绳，滑移块上移，凸轮被凹槽挡至垂直状，处于导轨外两侧，于是滑移座与下承梁分离，向上趋接上承梁，抓斗在离开地面瞬间由于重量作用，斗瓣打开卸料⟶卸完料的抓斗在空中处于打开状态，放在物料上面，滑移块下降，凸轮被顶针顶成约30°角，完成一个抓取过程⟶停工和休息时，不应将抓斗悬在空中。
6	The Grab Crane Stops(4311-GCR-001/002) 停止抓斗起重机(4311-GCR-001/002)	1. Orderly shutdown Anchanchor the crane in the designated position ⟶ Put the grab in the specified position, all control buttons should be put to the zero position, and the power switch of the electrical controlled lifting machinery should be disconnected⟶Clean the site for hygiene⟶ Lock the driver's cab. 1. 正常停车 将起重机停放在指定位置有效锚定⟶将抓斗收放到规定位置，所有控制按钮均应放到零位，电气控制的起重机械的电源开关应断开⟶打扫现场卫生⟶离开司机室时司机室应锁门。 2. Emergency Stop

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№	Stage	Key Points
		a. In an emergency, turn off the drive power; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, the operation shall be restarted in the driving sequence. c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭驱动电源； b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作。
7	Permits & isolation sheets 许可证和隔离表	1. Only authorised isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Grab Crane 抓斗起重机 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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Appendix

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Standard Operating Procedure
标准操作程序

Grab crane posts in the chloride plant
CP 抓斗起重机岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
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Approvals 批准

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